

TEZPUR UNIVERSITY

Spring Semester End-term Examination, 2018

CO205: Formal Languages & Automata (B.Tech)

Full Marks : 60

Time : 2 ¹/₂ hours

Figures on the right indicate marks for the individual question

1. Choose the correct option for the following questions. Correct answers will be awarded **+2** and incorrect answers will be awarded **-0.5**. [2×5=10]

(a) Which of the following statements are true for the language represented by the regular expression: $(a + b)^*a(a + b)^*a(a + b)^*$

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| (i) All strings have at most two a | (iii) No string can have two consecutive a |
| (ii) All strings have at least two a | (iv) No string can have more than three b |

(b) Which of the following regular expressions do NOT allow any string that ends with 101?

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|-----------------------|-----------------------------|
| (i) $1^*0^*1^*$ | (iii) $1^*0^*1(0 + 1)$ |
| (ii) $00^*(0 + 1)^*1$ | (iv) $(0 + 1)1^*(0 + 1)1^*$ |

(c) Which of the following statements are FALSE with respect to regular languages?

- (i) If a finite set of strings does not satisfy the Pumping Lemma, it is non-regular
- (ii) For a given regular language, the DFA with minimum number of states is unique
- (iii) In a DFA, ε -transitions are not allowed
- (iv) Two finite automata are said to be equivalent if they recognize the same language

(d) Which of the following is a regular language?

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| (i) $L = \{a^n b^m \mid n \geq 0, m \geq 0\}$ | (iii) $L = \{a^n b^m \mid n \text{ is odd and } m \text{ is even}\}$ |
| (ii) $L = \{a^n b^m \mid m = 2n, m, n \geq 0\}$ | (iv) $L = \{a^n b^m \mid n \geq m \geq 0\}$ |

(e) Given the CFG, $G = (\{S\}, \{a, b\}, P, S)$ where $P = \{S \rightarrow SS \mid aS \mid Sa \mid b\}$. Which of the following sets of strings can be generated from G ?

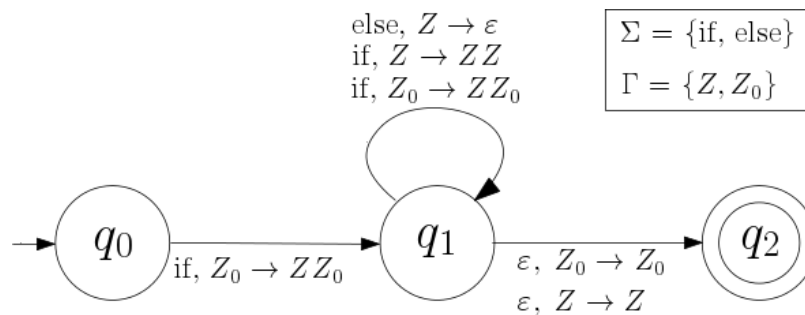
- | | |
|--------------------------------|-------------------------|
| (i) $\{aa, bb\}$ | (iii) $\{abba, baab\}$ |
| (ii) $\{\varepsilon, abaaba\}$ | (iv) $\{aabbba, baab\}$ |

2. (a) Construct a Context Free Grammar for the language L_1 .

$$L_1 = \{w \mid w \in \{a, b\}^*, |w| = 2n, n \geq 0\} \quad [5]$$

(b) What are the steps for converting a CFG to Chomsky Normal Form? [7]

3. Give the formal definition of Context Sensitive Grammars (CSG). Briefly state how CSGs are different from CFGs. [5+3]
4. Given the following transition diagram of a PDA, answer the questions listed below.



- (a) Describe in your own words the type of strings accepted by the PDA. [2]
- (b) Determine if the following strings are accepted by the PDA. Show the sequence of *instantaneous descriptions*.
 (i) if else if else else (ii) if else if [5+5]
5. Give the formal definition of Turing Machine. State the Church-Turing thesis. [5+3]
6. Answer **any two** of the following: [2×5 = 10]
- (a) What is the difference between *recursively enumerable* and *recursive* languages?
- (b) State the Halting Problem.
- (c) Using TMs, explain the difference between P and NP.